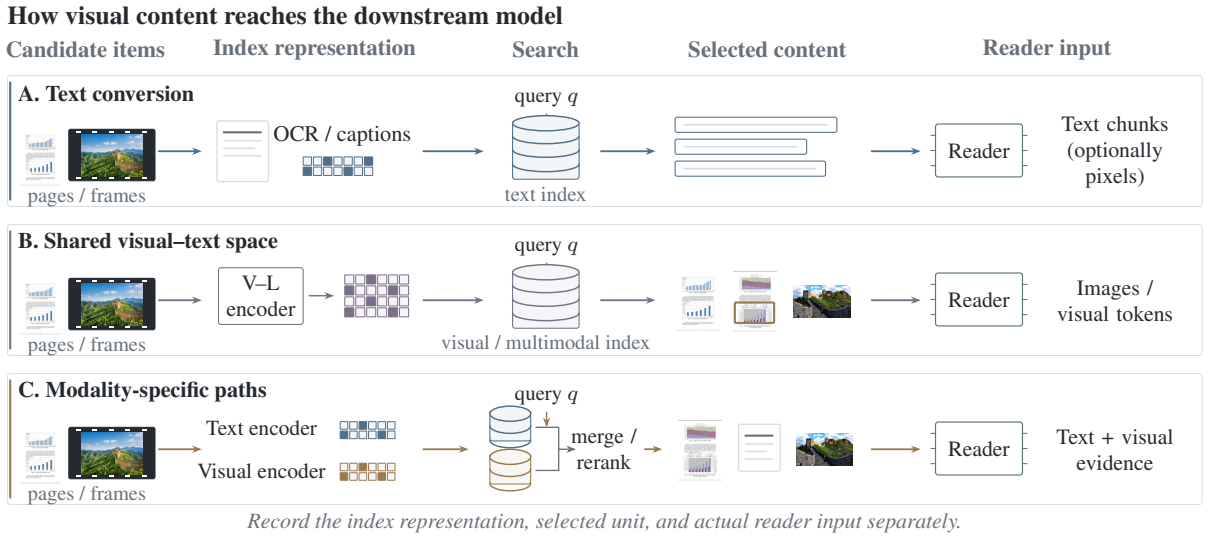

How visual content reaches the downstream model
Candidate items
Index representation
Search
Selected content
Reader input
A. Text conversion
pages / frames
OCR / captions
query q
text index
Reader
Text chunks (optionally pixels)
B. Shared visual–text space
pages / frames
V–L encoder
query q
visual / multimodal index
Reader
Images / visual tokens
C. Modality-specific paths
pages / frames
Text encoder
Visual encoder
query q
merge / rerank
Reader
Text + visual evidence
Record the index representation, selected unit, and actual reader input separately.